



Sound insulation of double-leaf walls – Allowing for studies of finite stiffness in a transfer matrix scheme

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Sound insulation of double-leaf walls – Allowing for studs of finite stiffness in a transfer matrix scheme

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ARTICLE INFO

Article history:

Received 9 December 2009
Received in revised form 3 February 2010
Accepted 4 February 2010
Available online 1 March 2010

Keywords:

Sound insulation
Sound radiation
Transfer matrix technique
Structural couplings
Double-leaf partitions

ABSTRACT

In a recent paper by the present author the effect of finite structural connections on the sound reduction index of double walls was predicted by modifying a model based on the transfer matrix technique. However, the model did not include any means to account for the flexibility of the studs; they were assumed to be of infinite stiffness. Based on data for the effective stiffness of flexible steel studs, the model is extended to take account of this flexibility. A number of comparisons are performed, mainly with the measured sound reduction index of lightweight double walls with gypsum boards. Cases include walls with cavity filling as well as with empty (air-filled) cavities. In the latter cases, the energy losses of the cavity are simulated using a model of a porous layer with a minute flow resistivity. Predicted results compare favourably with measurement results. It is assumed that different basic types of studs, i.e. other than the TC-type simulated here may successfully be included in the model.

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1. Introduction

In a recent paper by Vigran [1], reviving the semi-empirical method by Sharp [2] to account for the effect of structural connections, such as studs or ties, in double walls, the prediction method was based on a transfer matrix scheme. However, only connections of infinite stiffness were accounted for, which for studs in practice implies wooden ones only and not commonly used metal studs of various cross shapes.

It is well known that a large number of prediction methods exists in the literature and Hongisto [3] implemented all together seventeen of these models making a quantitative comparison between models against measurement results taken from a large series of laboratory measurements; see Hongisto et al. [4]. In the type of constructions we shall be concerned with here, double-leaf with structural connections a couple of models were able to predict the effect of flexible studs. However, only the model of Davy [5] gave satisfactory result, a model which has recently been somewhat modified; see Ref. [6]. The model compares favourably with a number of measurement results on plasterboard walls. However, it is depending on a reasonable guess on the size of a so-called stud attenuation factor, which must be based on comparison with measured end results, i.e. the sound reduction index of the walls.

More recently, Legault and Atalla [7] have, using a special double wall system consisting of two aluminium plates with a cavity filled with a fibrous material, compared altogether five models

including their own periodic model. Predictions are compared with measurements results in two cases; with C-sections channels connecting the aluminium plates as well as without these connections. Again, Davy's approach gives quite good agreement but cannot, in contrast to the periodic model, predict the finer details in the measured reduction index.

The role of studs in the sound transmission has recently been given broad coverage also by Poblet-Puig et al. [8], outlining procedures to obtain more accurate parameters to be used in prediction models. By numerical modelling, metal studs of altogether six different cross sections were tested in three types of lightweight double walls. The vibration level difference of the two leaves is a useful parameter for testing the effect of different studs on the same wall. However, as it will be dependant also on other variables, such as the material properties of the leaves, an averaged response corrected stiffness of the studs is given, presented as a frequency-dependant translational stiffness of the different types of studs. As opposed to this simulation procedure, one may certainly envisage a direct measurement in line with other methods on resilient elements; see e.g. Ref. [9].

In this paper, however, we shall use the stiffness data of Poblet-Puig et al. [8] to predict the sound reduction index of double walls with flexible studs, modifying the method given in Ref. [1] to account for the finite stiffness of the studs. For completeness, some important aspects of the method are repeated below.

For comparison with measurement we shall use data for double-leaf gypsum plasterboard walls in the cases where the flexibility data from Ref. [8] seems appropriate, i.e. when the dimensions of the studs are comparable. However, the special construction

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doi:10.1016/j.apacoust.2010.02.003

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used by Legault and Atalla [7] will also be modelled using the present approach.

2. Theory

Following Sharp [2], the diffuse field sound reduction index R of a double-leaf partition with sound bridges (see Fig. 1a), may be expressed as:

$$R = 10 \lg \frac{W_i}{W_{2,P} + W_{2,B}} \tag{1}$$

Here W_i is the incident power and the power radiated from plate 2 is divided into two parts, $W_{2,P}$ and $W_{2,B}$, the power radiated from plate 2 without the bridges and the power radiated due to the action of the bridges, respectively. These two contributions to the radiated power are assumed to be uncorrelated. Eq. (1) may be written as:

$$R = R_p - \Delta R = R_p - 10 \lg \left(1 + \frac{W_{2,B}}{W_{2,P}} \right) \tag{2}$$

where R_p is the sound reduction index of the double-leaf partition without the bridges and ΔR is the correction term due to the bridges. It may be shown that the power ratio in the correction term may be expressed as:

$$\frac{W_{2,B}}{W_{2,P}} = n \sigma_B \cdot \frac{v_B^2}{v_1^2} \cdot \left\langle \frac{v_1^2}{v_2^2} \right\rangle = n \sigma_B \cdot \frac{Z_{B1}^2}{Z_{B1} + Z_{B2}} \cdot \left\langle \frac{v_1^2}{v_2^2} \right\rangle \tag{3}$$

where σ_B is the radiation factor of the second plate driven by one of a number n bridges acting over the partition area S . The second term contains the input impedances of the plates seen from the sound bridge, which is given below. Expressions for the radiation factor, both for line- and point-driven plates are given in Ref. [1] and not repeated here.

The last term in Eq. (3) is the squared ratio of the velocities of plates 1 and 2 in the absence of the bridges assuming diffuse sound incidence, the latter indicated by the outer brackets $\langle \rangle$ Sharp [2] gives an approximate expression for this ratio but in the framework of transfer matrices there is no need to do so. For a given angle of incidence we may express this ratio in terms of the resulting transfer matrix comprising the construction without the bridges, i.e. two plates with a cavity partly or wholly filled with a porous layer. Using a thin plate model and representing any porous material as an equivalent fluid we get a 2×2 matrix with components A_{11} , A_{12} , A_{21} and A_{22} , see e.g. Refs. [3,10]. Denoting the squared velocity ratio at a given angle of incidence by r_θ , we get

$$r_\theta = \frac{v_1^2}{v_2^2} = |A_{21} \cdot Z_L + A_{22}|^2 \tag{4}$$

where Z_L represents the fluid loading impedance of the surrounding medium (air). Averaging this expression over the incidence angle θ

in the same way as when averaging the transmission coefficient to obtain R_p , will give us the last term in Eq. (3). In all examples shown here a 32-point Gaussian integration routine is used with an upper limit of 90°.

Sharp [2] assumed that the bridges were massless and infinitely stiff, which implies e.g. that the studs shown in Fig. 1a should be represented by massive wooden studs and not by the ones illustrated, which is a common shape of metal studs, denoted TC in Ref. [8]. To allow for flexible studs, we may as a thought experiment illustrated in Fig. 1b, simply insert a spring of mechanical impedance Z_s between the stud and plate 1. This will modify the ratio between the velocity v_B of the bridge and the velocity v_1 of plate 1, which for infinitely stiff bridges is given by:

$$\frac{v_B}{v_1} = \frac{Z_{B1}}{Z_{B1} + Z_{B2}} \tag{5}$$

being represented by the second term in Eq. (3). Introducing the spring of impedance Z_s we get a parallel combination with the impedance Z_{B1} of the input plate resulting in

$$\frac{v_B}{v_1} = \frac{Z_{B1}}{Z_{B1} + Z_{B2} \left(1 + \frac{Z_{B1}}{Z_s} \right)} = \frac{Z_{B1}}{Z_{B1} + Z_{B2} \left(1 - i\omega \frac{Z_{B1}}{s(\omega)} \right)} \tag{6}$$

where we have represented the impedance of the spring by a frequency-dependant translational stiffness s . As in Ref. [1] a time dependence $\exp(-i\omega t)$ is used. Furthermore, for the plate impedances we shall use the classical expressions for the input impedances of an infinite plate, which for point- and line-drive may be written:

$$\begin{aligned} Z_{B,\text{point}} &= 8\sqrt{mB} \\ Z_{B,\text{line}} &= \frac{2(1-i)m\omega}{k_p} \end{aligned} \tag{7}$$

where m , B and k_p are the plate mass per unit area, bending stiffness and free surface wavenumber, respectively.

It should be mentioned that other procedures to modify Sharp's expression to include flexible studs are given in the literature, e.g. Gu and Wang [11]. Their method is included in the comparison performed by Hongisto [3] as well as the one by Legault and Atalla [7].

3. Measured and predicted results

An extensive series of measured results on gypsum board walls is made available by the National Research Council of Canada, see Ref. [12]. Measurements comprise walls with wood studs and steel studs, staggered as well as directly coupling the boards. As for the steel studs, they appear to be of the type TC, mentioned above. Unfortunately, some material data is missing such as the elastic modulus of the boards used but this may be estimated from the surface weight and the coincidence frequency apparent from the measured data.

We shall use the measurement results for several of the walls with steel studs to compare with results using the prediction model outlined above. This necessitates an estimate of the translational stiffness s given in Eq. (6). As mentioned in the introduction we shall use the data given by Poblet-Puig et al. [8] to arrive at an estimate that may be used in our model. It should also be mentioned that data from Ref. [12] is also used by Davy [6] to compare with his model giving reasonably good agreement.

3.1. Stiffness of metal studs

From the calculated vibration level difference of the outer leaves of three different double walls, Poblet-Puig et al. [8] derived

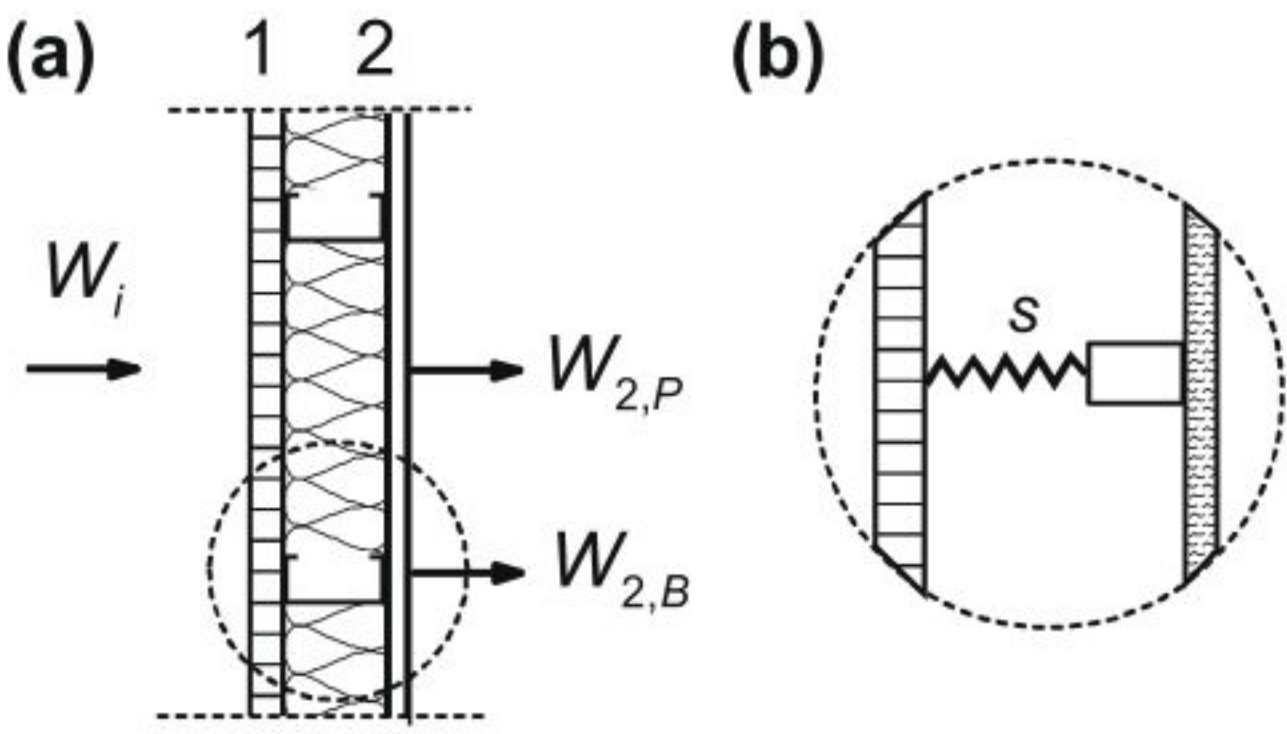
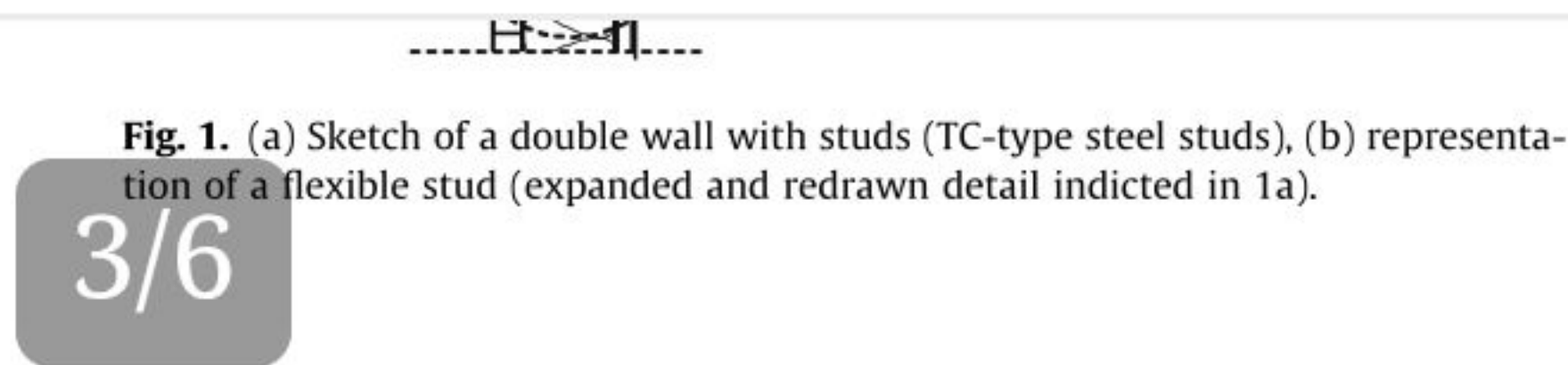


Fig. 1. (a) Sketch of a double wall with studs (TC-type steel studs), (b) representation of a flexible stud (expanded and redrawn detail indicated in 1a).



From the calculated vibration level difference of the outer leaves of three different double walls, Poblet-Puig et al. [8] derived

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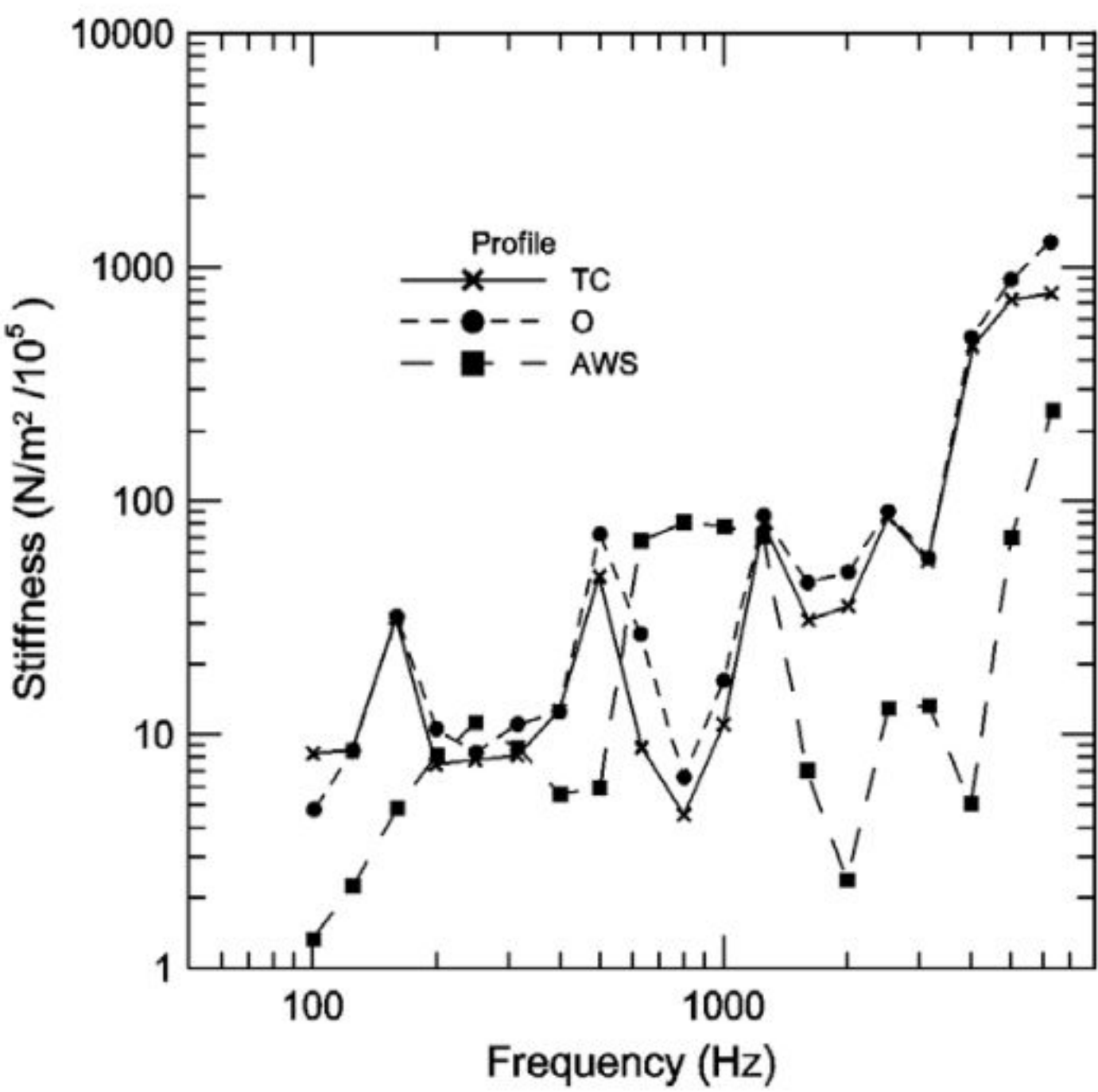


Fig. 2. Translational stiffness of three types of steel studs. Averaged values from numerical simulations on three different walls. Reproduced from Fig. 11 in Ref. [8].

the translational frequency dependant stiffness of altogether 5 different types of studs. The data is reproduced in Fig. 2 for three of these types, the mentioned TC-type, the O-type, which has a rectangular cross section (70 mm × 40 mm) and a special “acoustic” type AWS. The material thickness of all studs was 0.47 mm, with a largest cross dimension of 70 mm defining the cavity depth of the tested walls. As seen, the stiffness of the TC- and O-type is quite similar, whereas the AWS-type behaves differently. We shall use these data to obtain an approximate expression for the translational stiffness of the TC-type. To do so, we have plotted the data taking the logarithm of both the frequency and the stiffness, the result of which is shown in Fig. 3. Also shown by the dashed curve is a third-degree polynomial approximation to this curve, which we shall use in the comparisons below.

3.2. Cavity damping

Representing a porous material in the cavity between the leaves, we shall use an equivalent fluid model. As long as the stiffness of the material may be neglected and when the material is not glued to the leaves, the type of model used seems not to be critical. The problem arises when the cavity is empty, i.e. air-filled. There certainly will be energy losses in the cavity and how shall these be represented? A number of suggestions may be found in the literature such as using an absorption coefficient for the cavity of some 0.1–0.15 (Davy [6]), applying a loss factor η of 10^{-3} (Brunskog [13]) or a frequency independent power attenuation coefficient of $\approx 0.2 \text{ m}^{-1}$ (Vigran [14]). The latter was found expedient when applied to narrow air-filled cavities in double glazing’s but may not be suitable for cavities being 5–15 times wider. Also, it certainly should be expected that the absorption in the cavity should, at least slightly, increase with frequency.

As shown below, we have found that representing the air cavity by a porous material having a flow resistivity in the range 10–100 Pa s/m², is suitable for the lightweight double walls used here. The effective power attenuation coefficient is shown in Fig. 4 as a function of frequency with the flow resistivity as the parameter. Solid curves give data according to the relaxational model by Wilson [15], where the parameters are the ones giving the best fit to the model by Delany and Bazley [16], giving the dashed curves. It should be mentioned that the model by Wilson repairs the abnor-

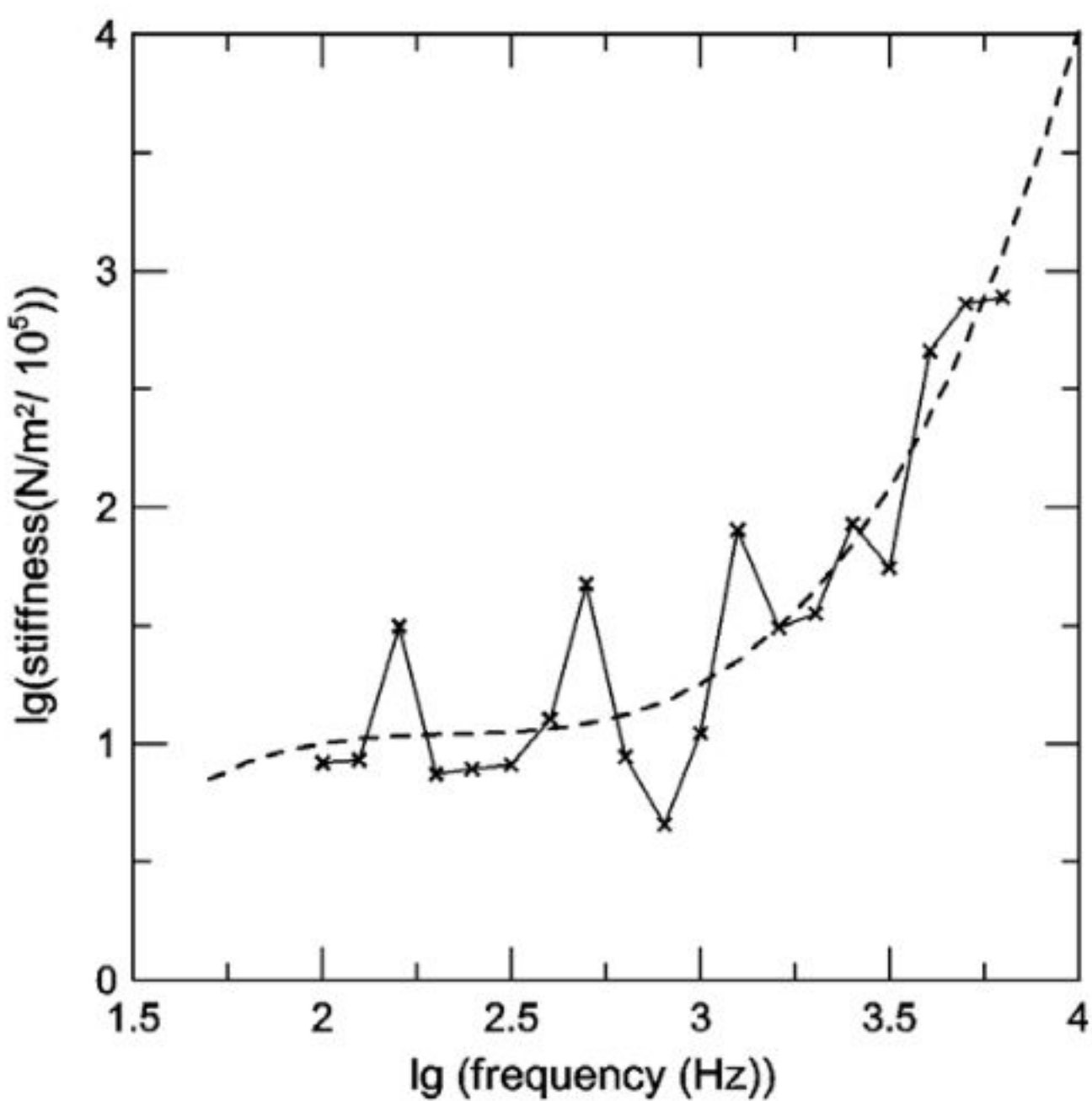


Fig. 3. The translational stiffness of the steel stud type TC represented by the logarithm of the stiffness. Dashed curve – the stiffness represented by a third-degree polynomial.

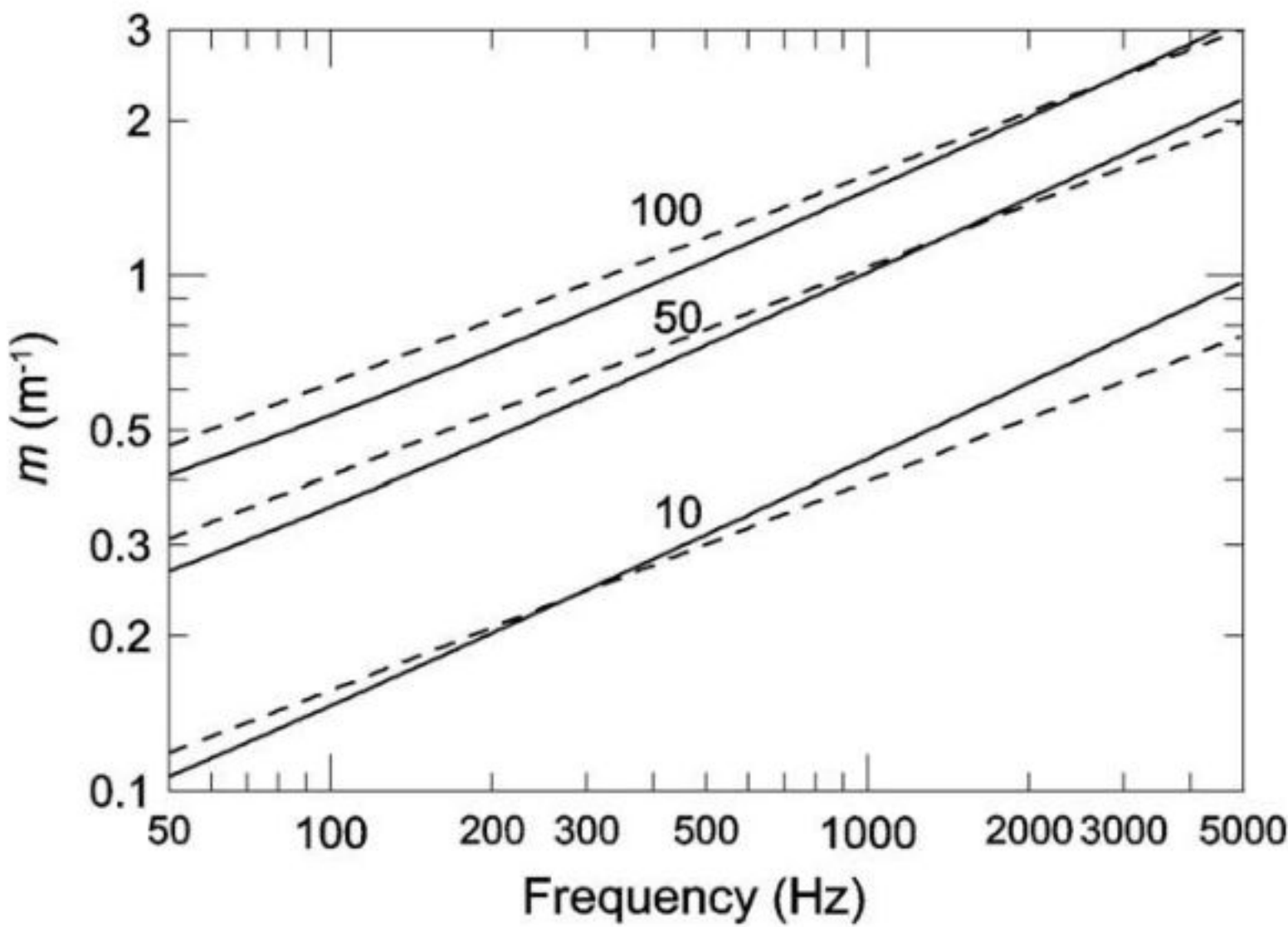


Fig. 4. Power attenuation coefficient of an equivalent fluid porous material, where the parameter is the flow resistivity in Pa s/m². Solid curves – Wilson model [15], dashed curves – Delany/Bazley-model [16].

mal behaviour of the Delany/Bazley-model at low frequencies but this has no relevance here.

3.3. Walls with metal studs

As mentioned above, most measured data used here are taken from the report given in Ref. [12], comprising measurement on double-leaf gypsum walls with steel studs of type denoted here as TC-type, the gypsum boards being nominally of thickness 13 and 16 mm, respectively. The material data used in the prediction model are given in Tables 1 and 2, for the gypsum boards and the cavity filling, respectively. The measured dimensions of the steel studs, used in the examples here were 32 mm × 65 mm, 30 mm × 91 mm and 30 mm × 150 mm, respectively. All studs had a material thickness of 0.53 mm, i.e. slightly thicker than the ones used in Ref. [8]. The wall test opening measured 3.05 m × 2.44 m, i.e. an area $\approx 7.5 \text{ m}^2$.

The first examples concern walls with cavity filling; a thick wall with a cavity depth of 150 mm and a thinner one of 65 mm cavity depth. Fig. 5 shows the measured results of these two walls to-





Table 1
Material characteristics of gypsum boards, Figs. 5, 6, 9 and 10.

Gypsum board	Thickness (mm)	Density (kg/m ³)	Modulus of elasticity (GPa)	Poisson's ratio	Loss factor (total)
I – 13 mm	12.7	790	2.5	0.3	0.06
II – 16 mm	15.9	690	2.5	0.3	0.06

Table 2
Material characteristics of porous material (cavity), Figs. 5, 6, 9 and 10.

Porous material	Type (marked in Ref. [12])	Flow resistivity (kPa s/m ²)	Thickness (mm)
A	Glass fibre (G1)	4.3	150
B	Mineral fibre (M1)	11.4	65
C	Air space	0.05	65, 90

Table 3
Weighted sound reduction index R_w .

Measurement no. TL	Figure no.	Measured R_w (dB)	Predicted R_w (dB)
93–299	5	52	53
93–064	5, 6	41	42
93–057	9	36	38
93–032	9	37	–
92–418	10	39	40
92–376	10	39	–

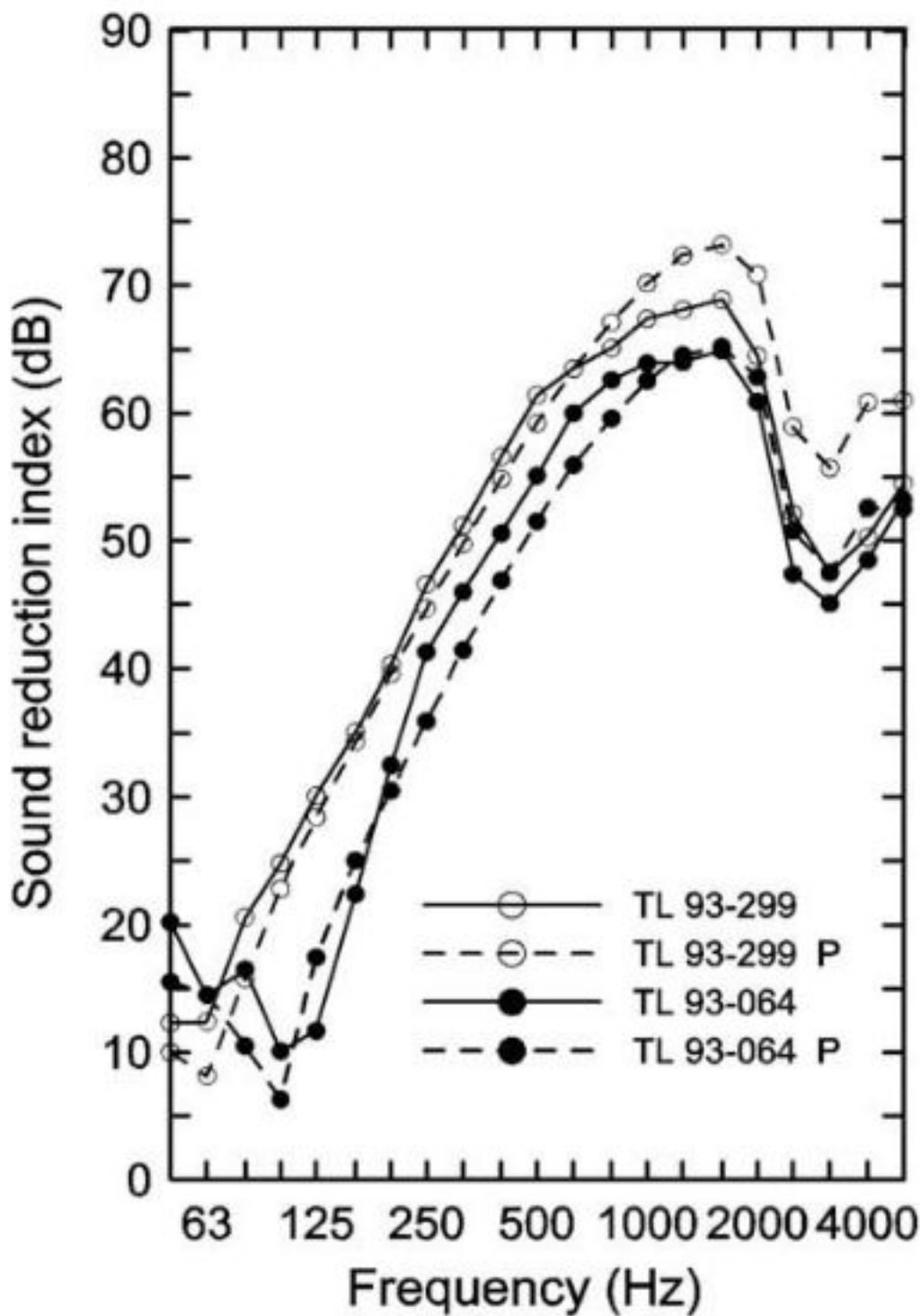


Fig. 5. Sound reduction index of double walls with cavity filling. The numbering is according to Ref. [12] and the capital letter P indicates predicted results. Material data according to Tables 1 and 2: TL 93-299 – (I, A), TL 93-064 – (I, B).

gether with the predicted results, the latter given by the dashed curves and indicated in the figure legend by the capital P. The TL-numbers refer to the original numbering in the report and the figure captions contains corresponding references to the material characteristics given in the tables, here (I, A) and (I, B). As the studs in the case of the 65 mm cavity depth have approximately the same dimensions as the ones with data given by Fig. 3, these data is directly applied when predicting the result of TL 93-064. For the case of TL 93-299, the cavity depth and thereby the longest cross dimension of the studs is more than twice the other. It is reasonably to assume that the effective stiffness is reduced and a factor of two is therefore used in this case. As seen, the fit between measured and predicted results is reasonably good in both cases. The weighted sound reduction index R_w calculated for the frequency range 100–3150 Hz is given in Table 3.

To illustrate the effect of the studs the curves for TL 93-064 are plotted again in Fig. 6 together with two simulated results; one giving the result assuming no structural connections, the other assuming studs of infinite stiffness, i.e. the latter result calculated using Eq. (3). As for the case of assuming studs of infinite stiffness, it is interesting to compare with the results given by Legault and Atalla [7]. The structure was here 1220 mm × 2030 mm alumin-

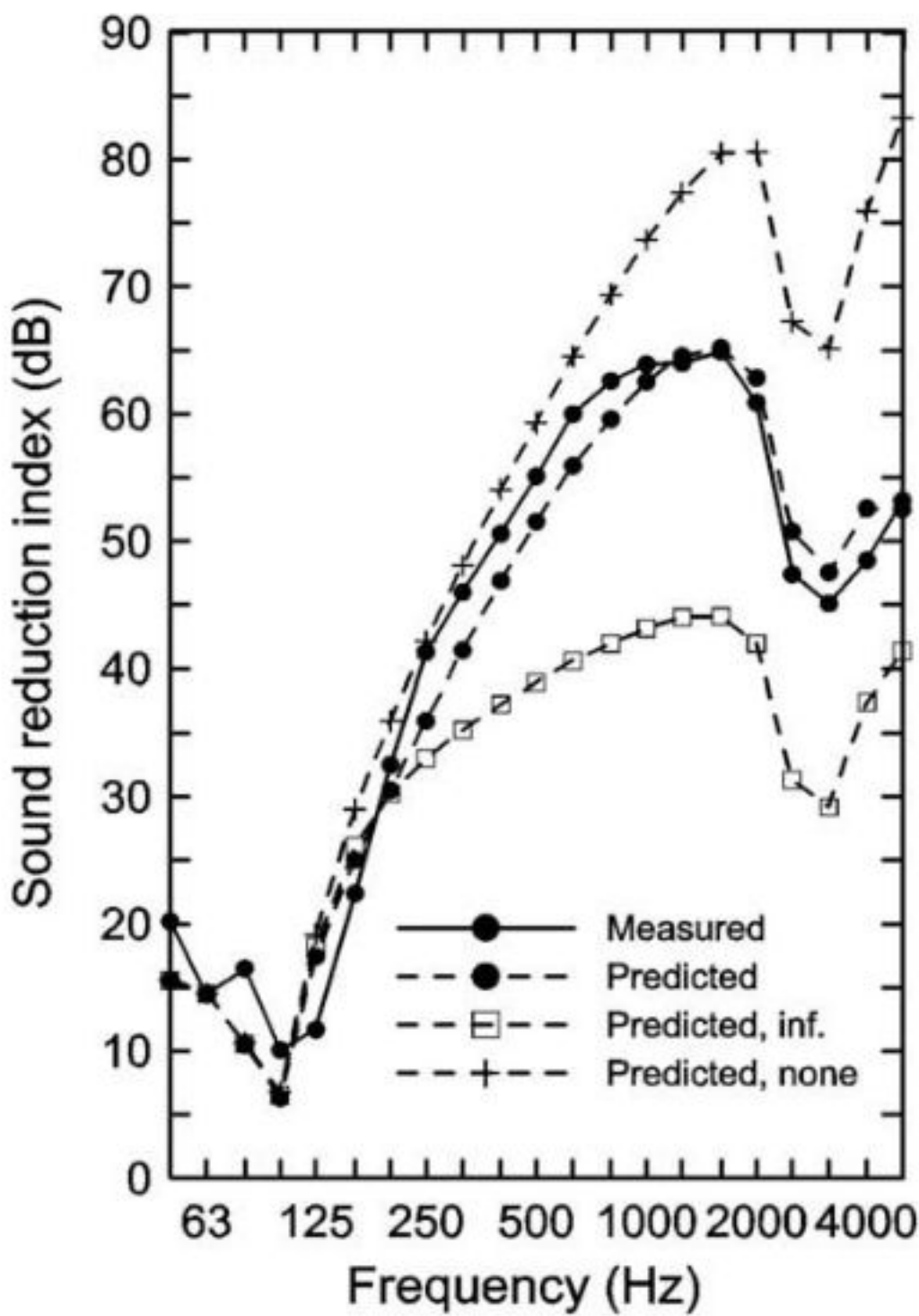


Fig. 6. Sound reduction index of double wall TL 93-064 (I, B), measured and predicted results. Simulated results are shown for the cases where the studs are of infinite stiffness as well as the same wall without studs.

ium plates (1 and 2 mm thick) with a cavity of 50.8 mm filled with a fibrous material. The material data for the plates and the Biot acoustic properties of the cavity filling are given in Tables 4 and 5. The “studs” are “C-section” aluminium channels with web length 50.8 mm (cavity depth) and spaced by a distance of 508 mm. As the measurements are performed in 1/24 octave bands and results averaged over 1/6 octave bands, we have in Figs. 7 and 8, reproducing the results by Legault and Atalla, retained a similar aspect ratio of the axis, i.e. different from the figures above.

Fig. 7 shows a comparison of the measured results with the predictions using the periodic model and the present model for the case of no connections between the plates, whereas Fig. 8 shows the coupled case. The present model is here using an Allard–Johnson model for the fibrous material; see Allard [17]. Furthermore, the C-channels coupling the plates have a material thickness of 3.175 mm, i.e. at least six times more solid than the examples above. We have therefore modelled these as being infinitely stiff which in fact gives quite good agreement with the measured result. However, such a simple model cannot be expected to predict any finer details in the result.





Atalla [7]. The structure was here 1220 mm × 2030 mm alumin-

any finer details in the result.

Table 4
Material characteristics of aluminium plates, Figs. 7 and 8.

Al plates	Density (kg/m ³)	Modulus of elasticity (GPa)	Poisson's ratio	Loss factor total
1 and 2 mm thick	2700	70	0.33	0.05

Table 5
Material characteristics of fibrous material, Figs. 7 and 8.

Flow resistivity (kPa s/m ²)	17.7
Porosity	0.91
Tortuosity	1.0
Viscous length (m)	128·10 ^{−6}
Thermal length (m)	376·10 ^{−6}

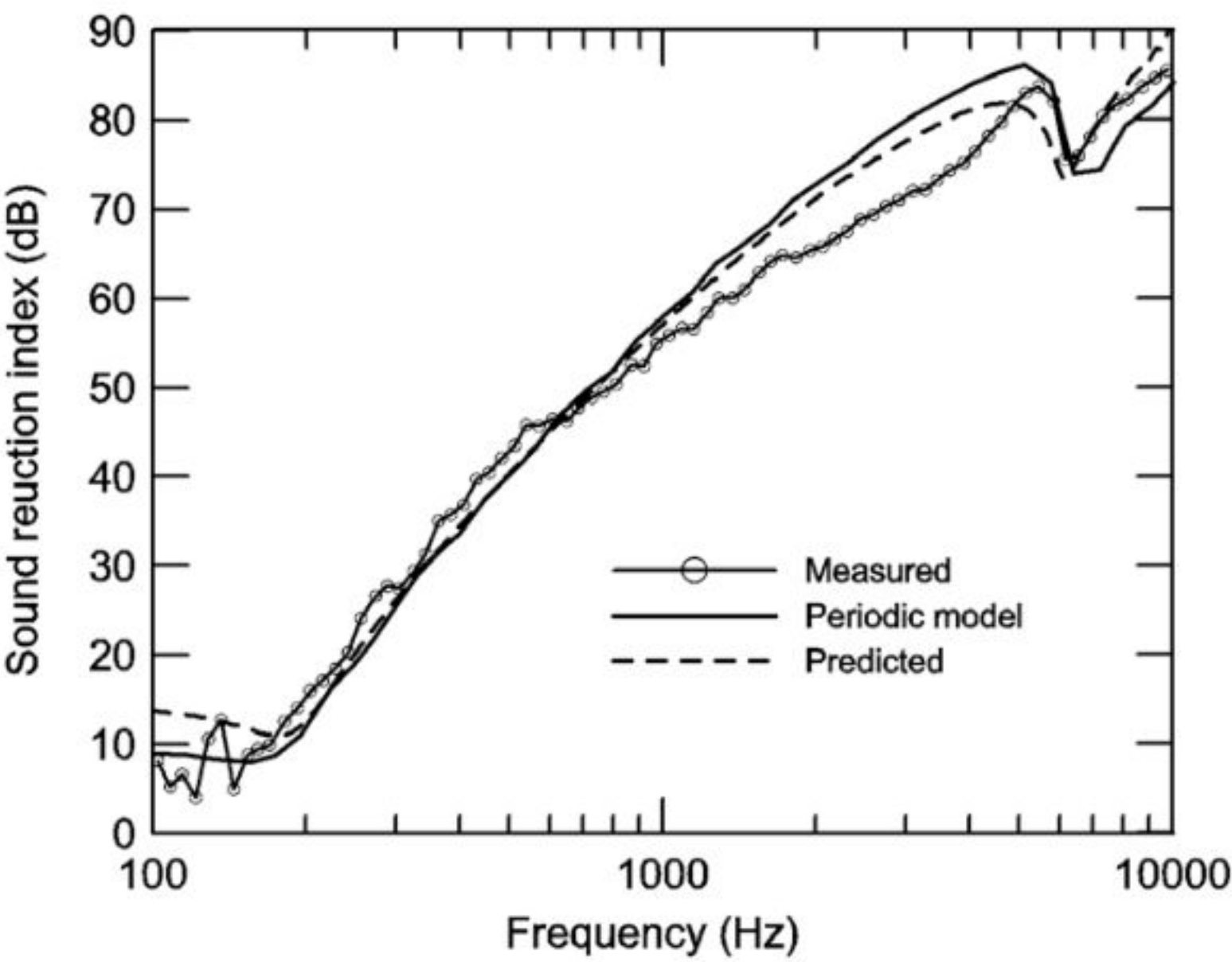


Fig. 7. Sound reduction index with no structural connections between aluminium plates. Measured and predicted results with periodic model reproduced from Fig. 6 in Ref. [7], together with prediction using the present model. For material data, see Tables 4 and 5.

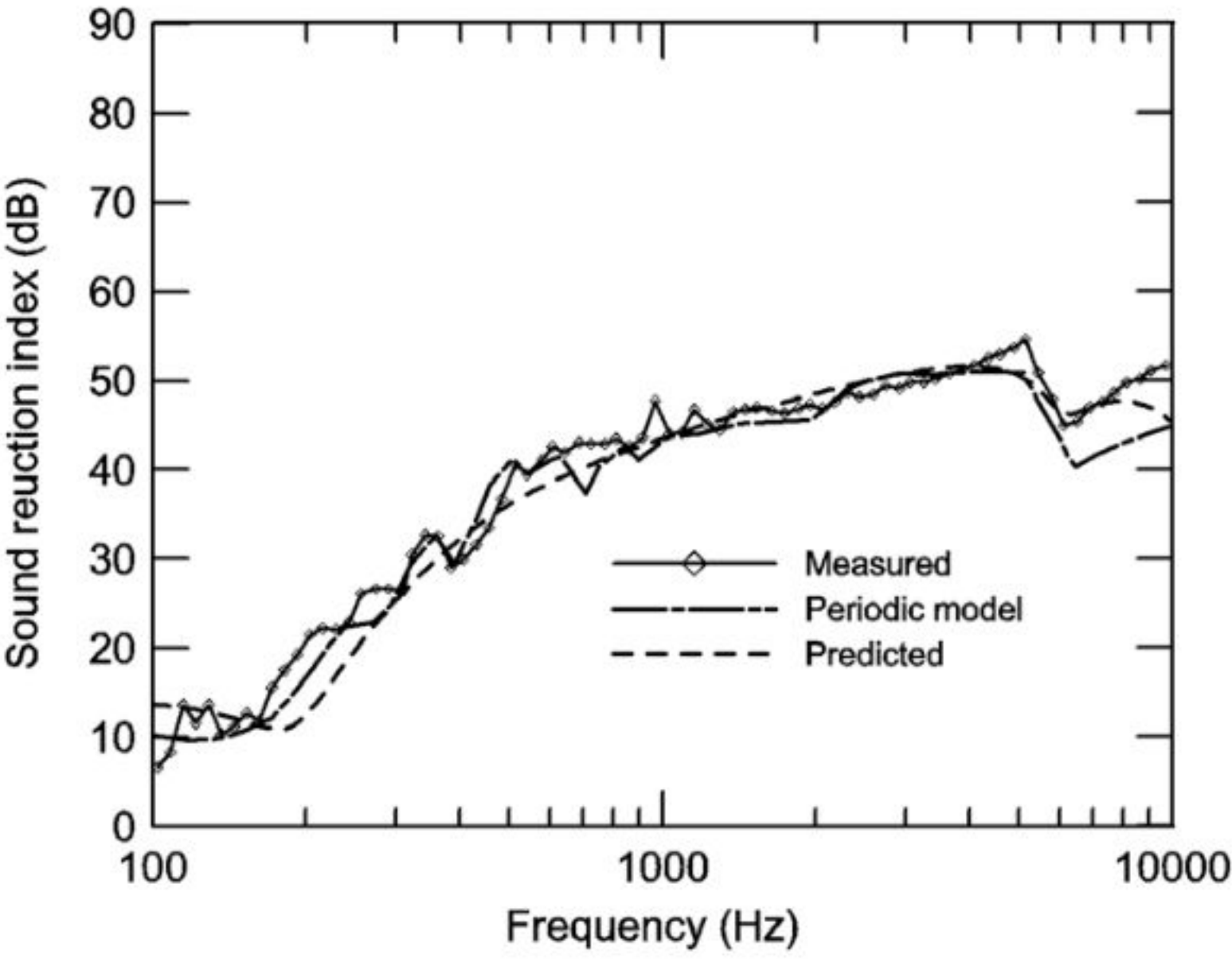


Fig. 8. Sound reduction index with C-section channels between aluminium plates. Measured and predicted results with periodic model (non-rigid and massless beam case) reproduced from Fig. 6 in Ref. [7], together with prediction using the present model. For material data, see Tables 4 and 5.

In the last two figures, Figs. 9 and 10, the measured data are again taken from Ref. [12] concerning gypsum board walls. The

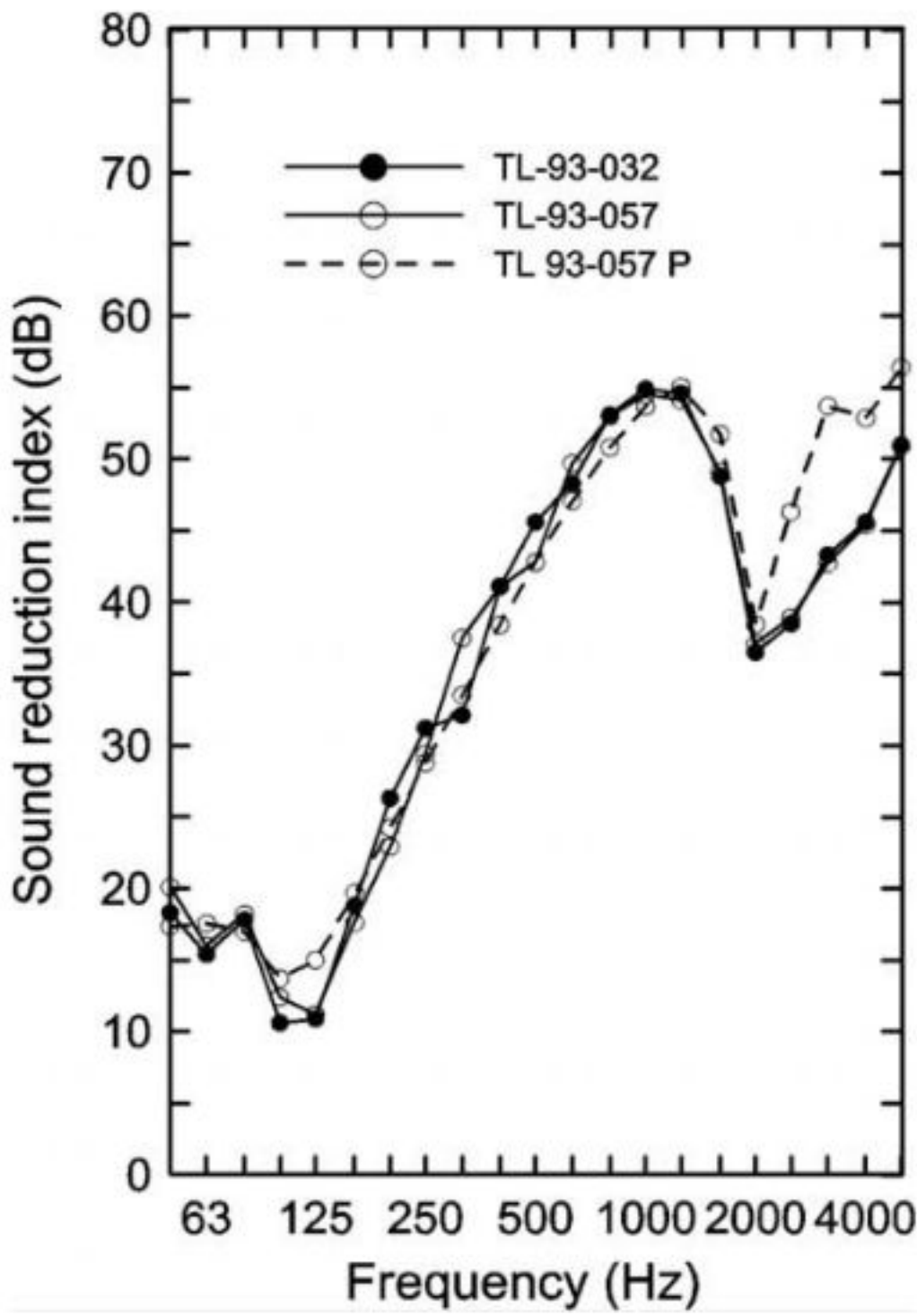


Fig. 9. Sound reduction of double walls with steel studs and an empty cavity of depth 65 mm. The stud spacing is 406 mm (TL 93-057) and 610 mm (TL 93-032), respectively. Predicted result is indicated by the capital letter P. Material data (II, C), see Tables 1 and 2.

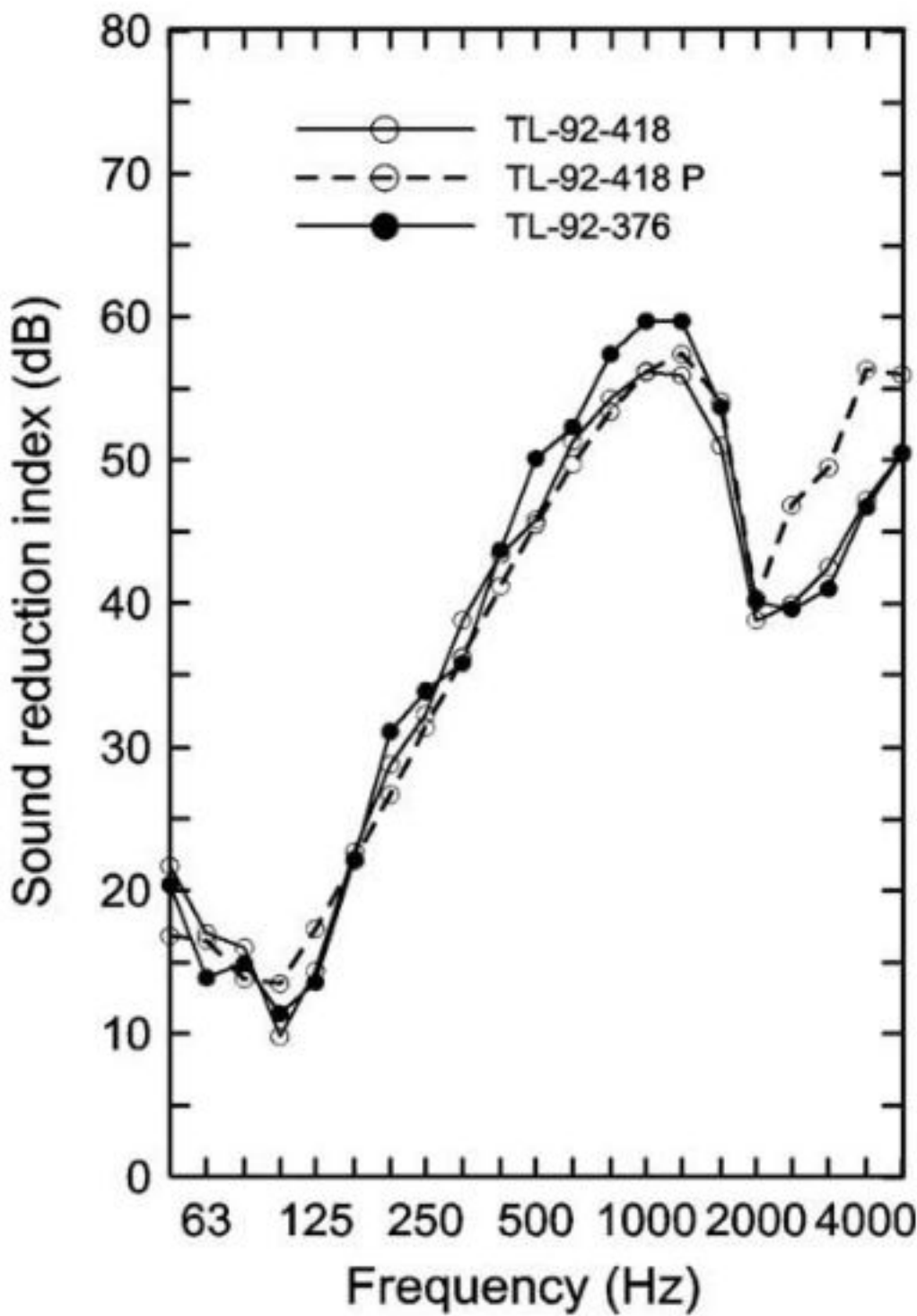


Fig. 10. Sound reduction of double walls with steel studs and an empty cavity of depth 90 mm. The stud spacing is 406 mm (TL-92-418) and 610 mm (TL-92-376), respectively. Predicted result is indicated by the capital letter P. Material data (II, C), see Tables 1 and 2.

cavity is empty and the energy losses in the cavity is simulated by a porous medium having a flow resistivity of 50 Pa s/m², giving an effective power attenuation coefficient of the air-filled cavity as shown in Fig. 4. In the examples shown in Fig. 9 the gypsum boards were of nominal thickness 16 mm fastened to steel studs of 65 mm thickness. Two measured results are shown, where the only difference is the stud spacing being 406 mm and 610 mm, respectively. As seen the difference between the two curves is quite small, with a difference in the weighted sound reduction index of one dB; see Table 3. In the predicted result a stud spacing of 610 mm is used.

Similar results are presented in Fig. 10, where the only difference from the ones given in Fig. 9 is that the stud thickness is in-





creased to 90 mm. However, both in Figs. 9 and 10, the predicted results are calculated using directly the stiffness curve presented in Fig. 3. In both examples the difference between measured and predicted results is most noticeable above the critical frequency, probably due to insufficient prediction of the resonant part of the transmission.

4. Conclusions

To account for the effect of structural connections (sound bridges) in double walls the semi-empirical method proposed by Sharp [2] was revived by Vigran [1] and used to correct resulting data from the transfer matrix method calculation on the construction without bridges. However, this was limited to cases assuming studs of infinite stiffness. In the present work, the method is extended to include flexible studs using data from Ref. [8] for the effective stiffness for one type of steel stud. A comparison with a number of measured results on the sound reduction index of lightweight double walls with gypsum boards is performed, including cases with and without cavity filling. Some recent results on a construction with aluminium plates are also included, in which case the bridges are assumed to be of infinite stiffness. Taking the simplifications of the method into account, such as neglecting the weight of the studs, the fit between measured and predicted results is quite good.

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